

A Unbiasedness of GPBM

Buchholz et al. [8] showed that, when the correct position bias curve is known, and the chosen window system satisfies a condition known as *common window support*³, the INTERPOL estimator is unbiased. We now prove a similar claim for GPBM using similar assumptions. Since GPBM uses weights (F) instead of windows, we require an analog of common window support, adapted for the weight matrix, F .

Definition A.1. For a given target policy, π , and weight matrix, F , we say that a logging policy, π_0 , has *common support with π under F* if, for every item $d \in \mathcal{D}$ and target position $j \in \{1, \dots, K\}$ where $\pi(d, j) > 0$, exists a logging position $k \in \{1, \dots, K\}$ such that $(F(j, k) \cdot \hat{p}_k > 0) \wedge (\pi_0(d, k) > 0)$.

Definition A.1 can be interpreted as saying that, if π ranks an item d at position j , then π_0 must have nonzero probability of placing d in at least one position k where the product $F(j, k) \cdot \hat{p}_k > 0$. Note that, when the entries of F are binary (and we recover INTERPOL), this condition is equivalent to common window support.

We are now ready to state our main result.

PROPOSITION A.2. Fix an i.i.d. dataset, D_N , consisting of rankings generated by a logging policy π_0 , and rewards generated under the PBM with positive position bias $p > 0$. Fix a position bias estimate, $\hat{p}$, and assume that $\hat{p} = p$. Then, for any target policy π , and weight matrix F , such that π_0 has common support with π under F (Definition A.1), we have that GPBM is an unbiased estimate of the expected reward of π .

PROOF. Similar to Buchholz et al.'s proof, it suffices to analyze the expectation of a single record in the dataset, due to independence and linearity of expectation.

$$\mathbb{E} \left[\sum_{d \in \mathcal{D}} \sum_{k=1}^K \sum_{j=1}^K \frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \mathbf{1}[y_k = d] c(y, k) \right] \quad (18)$$

$$= \sum_{d \in \mathcal{D}} \sum_{k=1}^K \sum_{j=1}^K \frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \pi_0(d, k) \mathbb{E}[c(y, k)] \quad (19)$$

$$= \sum_{d \in \mathcal{D}} \sum_{k=1}^K \sum_{j=1}^K \frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \pi_0(d, k) \theta_d p_k \quad (20)$$

$$= \sum_{d \in \mathcal{D}} \theta_d \sum_{j=1}^K \pi(d, j) \hat{p}_j \frac{\sum_{k=1}^K F(j, k) \cdot \pi_0(d, k) \cdot p_k}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \quad (21)$$

$$= \sum_{d \in \mathcal{D}} \theta_d \sum_{j=1}^K \pi(d, j) \hat{p}_j \quad (22)$$

$$= \Delta(\pi). \quad (23)$$

The assumptions of common support ensure that the denominator, $\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}$, is never zero. $\square$

B Additional Experimental Details

Following Buchholz et al. [8], we evaluate all models across a wide range of position bias misestimation patterns $\hat{p}$. However, in the

³In their paper, Buchholz et al. call this "full" window support; but given the dependence on the target policy, we feel it is more appropriate to call this "common" window support.

above settings, all positions $\hat{p}_k$ are misestimated in the same direction (all positions either over- or underestimated), and the deviation between the two curves changes monotonically (grows for $\hat{p} \in \{p^{0.6}, p^{0.8}, p^{1.2}, p^{1.4}\}$ or is fixed for $\hat{p} \in \{p^{1.0}, \max(p - 0.1, 0)\}$). We thus evaluate the models under a more complex misestimation pattern we term the *zigzag* setting $\hat{p} = p \pm z$. In this setting, the estimated curve $\hat{p}$ at position $k \in \{1, \dots, K\}$ is defined as:

$$\hat{p}_k = \min(1, p + (-1)^{(k+1)} \cdot 0.1 \cdot \sin(\frac{\pi \cdot k}{K+1})), \quad (24)$$

where $\pi \approx 3.14$ and $\sin$ is the sine function. Using this formulation, $\hat{p}$ is overestimated relative to p at odd positions, and underestimated at even positions. Furthermore, the magnitude of the deviation between the two curves progressively increases from the first rank until the center of the ranking, followed by a symmetric decay for the second part of the ranking.

C Gradient of $\hat{\Delta}_{\text{GPBM}}$ w.r.t. the Estimated Position Bias Curve

The gradient of the GPBM estimator $\hat{\Delta}_{\text{GPBM}}(\pi; \hat{p}, D_M^{(b)})$ using the estimated position bias curve $\hat{p}$ for a single bootstrap sample of ranking data $D_M^{(b)}$ w.r.t. $\hat{p}$ is:

$$\nabla_{\hat{p}} \hat{\Delta}_{\text{GPBM}}(\pi; \hat{p}, D_M^{(b)}) = \nabla_{\hat{p}} \left[\frac{1}{M} \sum_{i=1}^M \sum_{k=1}^K c(y^{(b,i)}, k) w(y_k^{(b,i)}, k) \right] \quad (25)$$

$$= \frac{1}{M} \sum_{i=1}^M \sum_{k=1}^K c(y^{(b,i)}, k) \nabla_{\hat{p}} w(y_k^{(b,i)}, k), \quad (26)$$

where $y^{(b,i)}$ the i 'th sampled ranking in the bootstrap sample $D_M^{(b)}$, $y_k^{(b,i)}$ is the item at the k 'th position of that ranking and $\nabla_{\hat{p}} w(y_k^{(b,i)}, k)$ is the vector of partial derivatives $[\frac{\partial w(y_k^{(b,i)}, k)}{\partial \hat{p}_1}, \dots, \frac{\partial w(y_k^{(b,i)}, k)}{\partial \hat{p}_K}]$ of individual weights w.r.t. each value in the estimated position bias curve $\hat{p}$. For a given weight $w(d, k)$, this partial derivative w.r.t. $\hat{p}_m$, the estimated position bias value at position m , is then equal to:

$$\frac{\partial w(d, k)}{\partial \hat{p}_m} = \frac{\partial}{\partial \hat{p}_m} \left[\sum_{j=1}^K \frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \right] \quad (27)$$

$$= \sum_{j=1}^K \frac{\partial}{\partial \hat{p}_m} \left[\frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \right] \quad (28)$$

$$= \sum_{j=1}^K \frac{\frac{\partial}{\partial \hat{p}_m} [F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j]}{\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} - F(j, k) \quad (29)$$

$$= \frac{\pi(d, j) \cdot \hat{p}_j \cdot \frac{\partial}{\partial \hat{p}_m} \left[\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'} \right]}{\left(\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'} \right)^2} \quad (30)$$

$$= \frac{F(m, k) \cdot \pi(d, m)}{\sum_{k'=1}^K F(m, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \quad (31)$$

$$- \sum_{j=1}^K \frac{F(j, k) \cdot \pi(d, j) \cdot \hat{p}_j \cdot F(j, m) \cdot \pi_0(d, m)}{\left(\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'} \right)^2} \quad (32)$$

$$= \frac{F(m, k) \cdot \pi(d, m)}{\sum_{k'=1}^K F(m, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \quad (33)$$

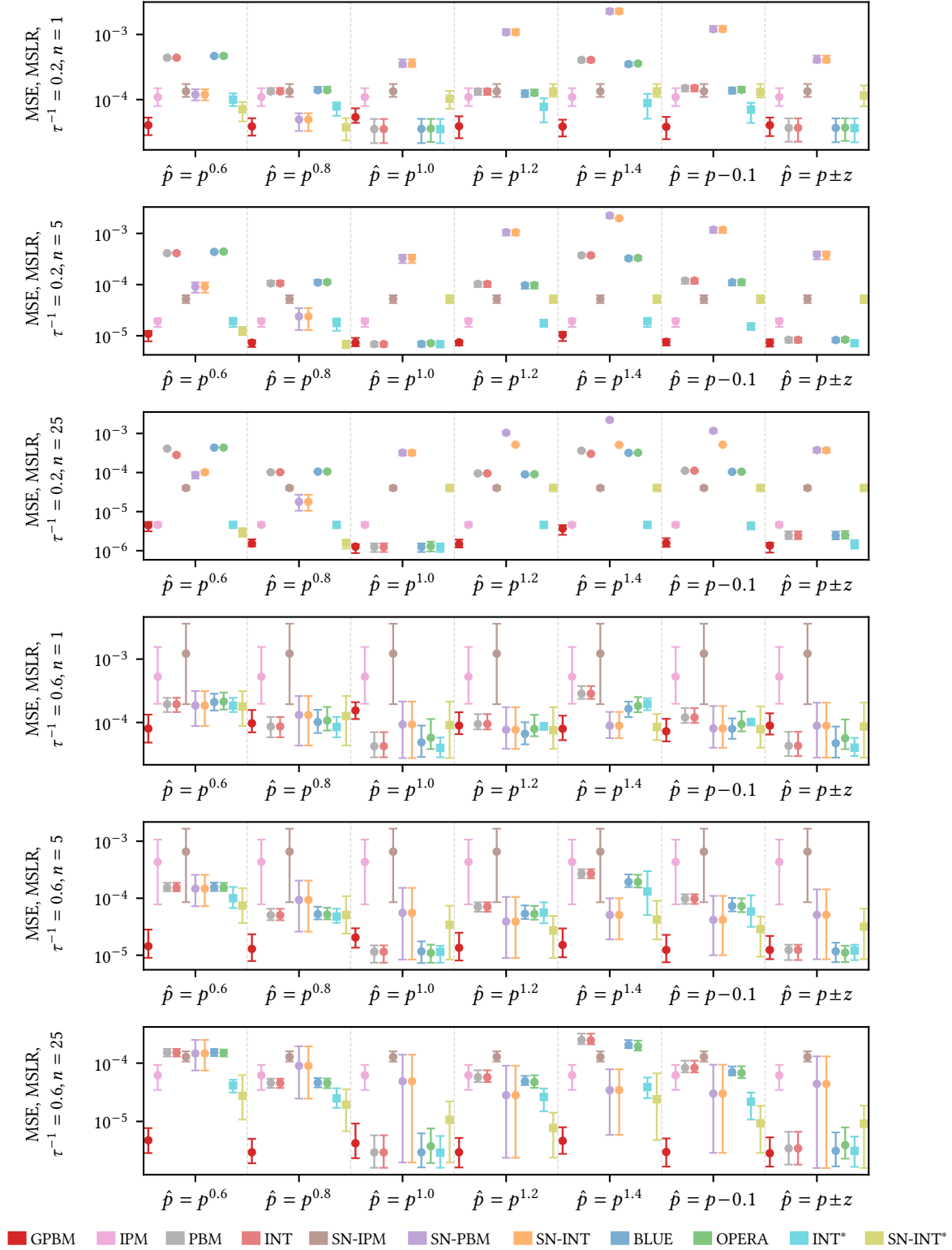
$$\cdot \left(1 - \frac{F(m, m) \cdot \pi_0(d, m) \cdot \hat{p}_m}{\sum_{k'=1}^K F(m, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'}} \right) \quad (34)$$

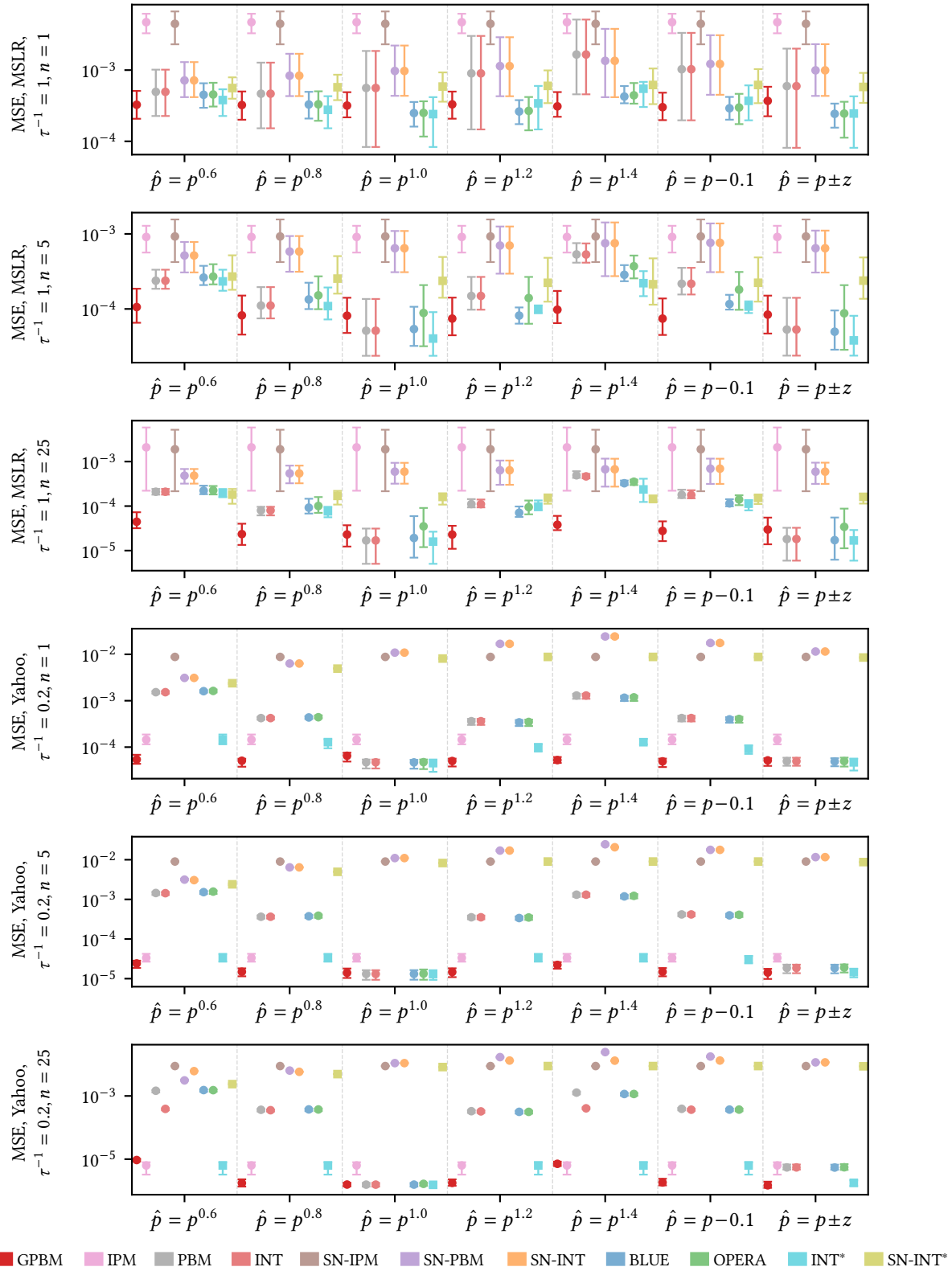
$$- \sum_{j \neq m} \frac{F(j, k) \cdot F(j, m) \cdot \pi(d, j) \cdot \pi_0(d, m) \cdot \hat{p}_j}{\left(\sum_{k'=1}^K F(j, k') \cdot \pi_0(d, k') \cdot \hat{p}_{k'} \right)^2}. \quad (35)$$

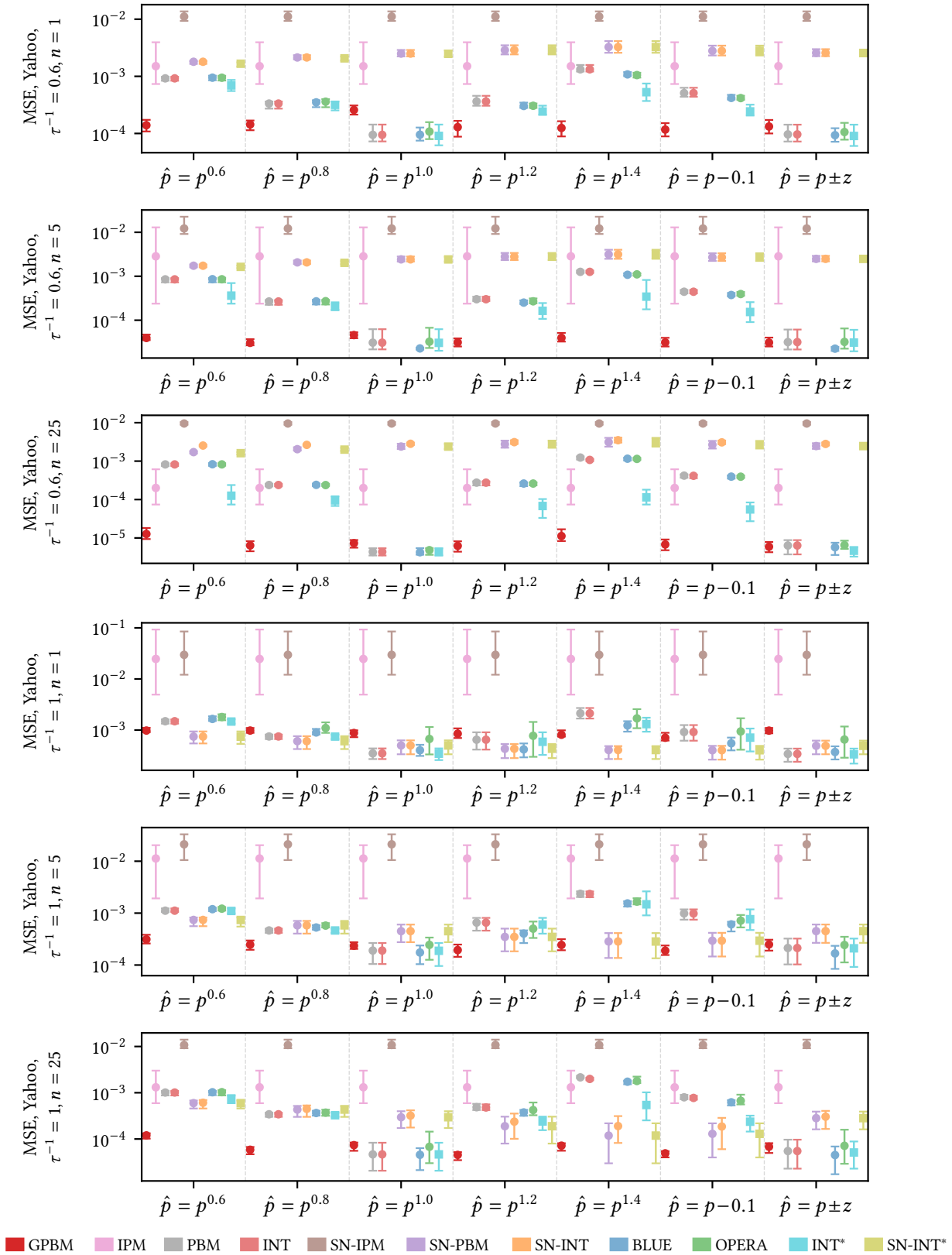
The full gradient $\nabla_{\hat{p}} \hat{\Delta}_{\text{GPBM}}(\pi; \hat{p}, D_M^{(b)})$ is then obtained by collecting the partial derivatives $\frac{\partial w(y_k^{(b,i)}, k)}{\partial \hat{p}_m}$ for all position bias curve positions m across all clicks in the bootstrap sample.

D Full Results

D.1 PBM User Behavior (RQ1)







D.2 IPM User Behavior (RQ2)

